10.1 To be the central atom in a compound, the atom must be able to simultaneously bond to at least two other atoms. He, F, and H cannot serve as central atoms in a Lewis structure. Helium is a noble gas, and as such, it does not need to bond to any other atoms. Hydrogen ($1s^1$) and fluorine ($1s^22s^22p^5$) only need one electron to complete their valence shells. Thus, they can only bond to one other atom, and it does not have *d* orbitals available to expand its valence shell.

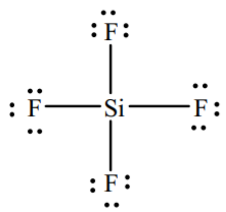
10.3 For an element to obey the octet rule it must be surrounded by 8 electrons. To determine the number of electrons present (1) count the individual electrons actually shown adjacent to a particular atom, and (2) add two times the number of bonds to that atom. Using this method the structures shown give: (a) 0 + 2(4) = 8; (b) 2 + 2(3) = 8; (c) 0 + 2(5) = 10; (d) 2 + 2(3) = 8; (e) 0 + 2(4) = 8; (f) 2 + 2(3) = 8; (g) 0 + 2(3) = 6; (h) 8 + 2(0) = 8. All the structures obey the octet rule except: c and g.

10.5 Plan: Count the valence electrons and draw Lewis structures.

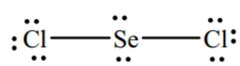
Solution:

Total valence electrons: SiF_4 has 32; $SeCl_2$ has 20; and COF_2 has 24. The Si, Se, and the C are the central atoms, because these are the elements in their respective compounds with the lower group number (in addition, we are told C is central). Place the other atoms around the central atoms and connect each to the central atom with a single bond. At this point the Si has an octet, and the remaining electrons are placed around the fluorines (3 pairs each). The 2 bonds plus 2 lone pairs complete the octet on Se and leave enough electrons to complete the Cl octet with 3 pairs each. The 3 bonds to the C leave it 2 electrons short of an octet. Forming a double bond to the O completes the C octet, and leaves sufficient electrons to form 2 lone pairs on the O and 3 lone pairs on each of the F's. All atoms in each of the structures end with an octet of electrons.

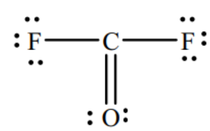
(a) SiF_4



(b) $SeCl_2$



(c) COF_2

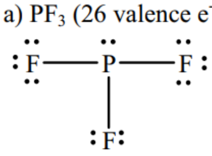


10.7 Plan: Count the valence electrons and draw Lewis structures.

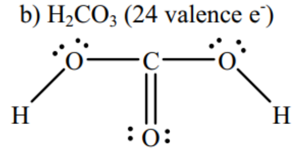
Solution:

The P and the C's are central. Two of the O's in part b also are central. Place appropriate single bonds between all atoms. The O in part b without a hydrogen needs a double bond as do both the S's in part c. The remaining electrons serve as lone pairs. All atoms, except the H's, have octets.

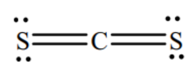
a) PF_3 (26 valence e^-)



b) H_2CO_3 (24 valence e^-)



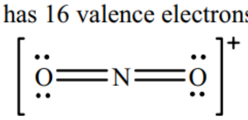
c) CS_2 (16 valence e^-)



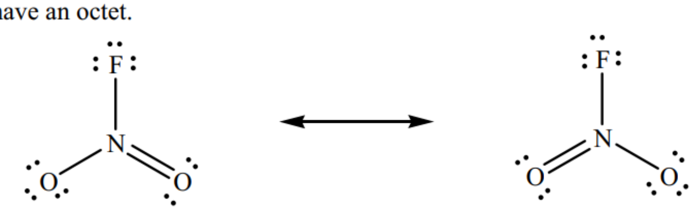
10.9 Plan: The problem asks for resonance structures, so there must be more than one answer for each part.

Solution:

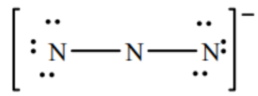
a) NO_2^+ has 16 valence electrons. No resonance is required as all atoms can achieve an octet with double bonds.



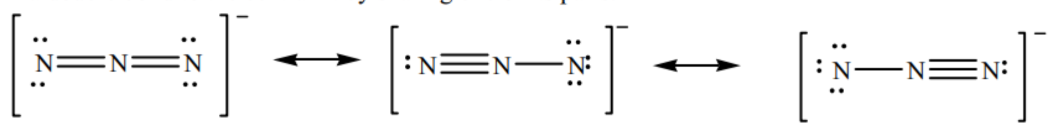
b) NO_2F is like NO_2, but has an extra atom (F) to interact with the lone electron on the N. In this structure all atoms have an octet.



10.11 Plan: Count the valence electrons and draw Lewis structures. Additional structures are needed to show resonance.
Solution:
a) N_3^- has 16 valence electrons.
The nitrogens must all be attached to each other with at least one bond. This uses 4 electrons leaving 12 electrons (6 pairs). Giving 3 pairs on each end nitrogen gives them an octet, but leaves the central N with only 4 electrons as shown below:

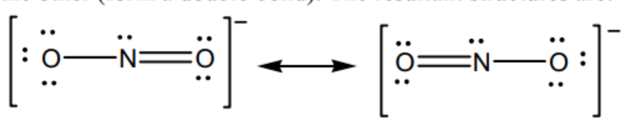


The central N needs 4 electrons. There are three options to do this: (1) one of the end N's could form a triple bond by sharing two of its lone pairs; (2) the other end N could form the triple bond instead; (3) each of the end N's could form a double bond to the central N by sharing one of its pairs.

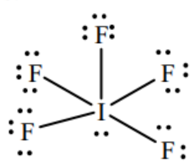


FC = -1 +1 -1 0 +1 -2 -2 +1 0
Formal charges (FC) are shown for comparison.

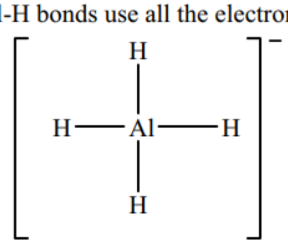
b) NO_2^- has 18 valence electrons.
The nitrogen should be the central atom with each of the oxygens attached to it by a single bond (2 x 2 = 4 electrons). This leaves 14 electrons (7 pairs). If 3 pairs are given to each O and 1 pair is given to the N, then both O's have an octet, but the N only has 6. To complete an octet the N needs to gain a pair of electrons from one O or the other (form a double bond). The resultant structures are:



10.13 Plan: Initially, the method used in the preceding problems may be used to establish a Lewis structure. The total of the formal charges must equal the charge on an ion or be equal to 0 for a compound. The FC only needs to be calculated once for a set of identical atoms.
Solution:
a) IF_5 has 42 valence electrons.
The presence of 5 F's around the central I means that the I will have a minimum of 10 electrons; thus, this is an exception to the octet rule. The 5 I-F bonds use 10 electrons leaving 32 electrons (16 pairs). Each F needs 3 pairs to complete an octet. The 5 F's use 15 of the 16 pairs, so there is 1 pair left for the central I.
This gives:



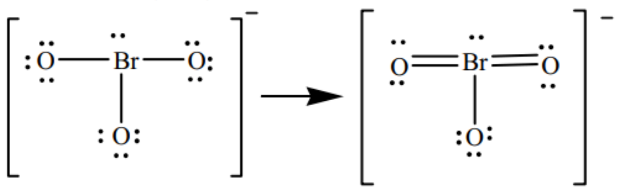
Calculating formal charges:
FC = valence electrons - (lone electrons + 1/2 (bonded electrons))
For iodine: $FC_I = 7 - (2 + 1/2(10)) = 0$
For each fluorine $FC_F = 7 - (6 + 1/2(2)) = 0$
Total formal charge = 0 = charge on the compound.
b) AlH_4^- has 8 valence electrons.
The 4 Al-H bonds use all the electrons and leave the Al with an octet.



For aluminum: $FC_{Al} = 3 - (0 + 1/2(8)) = -1$
For each hydrogen $FC_H = 1 - (0 + 1/2(2)) = 0$
Total formal charge = -1 = charge on the ion.

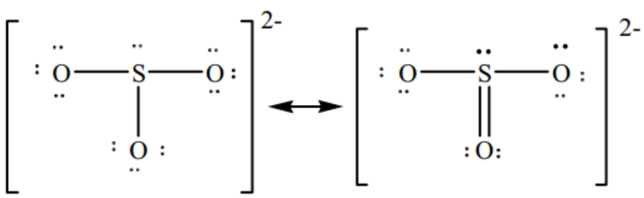
10.17 Plan: The general procedure is similar to the preceding problems, plus the oxidation number determination.
Solution:
a) BrO_3^- has 26 valence electrons.
Placing the O's around the central Br and forming 3 Br-O bonds uses 6 electrons and leaves 20 electrons (10 pairs). Placing 3 pairs on each O (3 x 3 = 9 total pairs) leaves 1 pair for the Br and yields structure I below. In structure I, all the atoms have a complete octet. Calculating formal charges:
$FC_{Br} = 7 - (2 + 1/2(6)) = +2$ $FC_O = 6 - (6 + 1/2(2)) = -1$
The FC_O is acceptable, but FC_{Br} is larger than is usually acceptable. Forming a double bond between any one of the O's gives:
$FC_{Br} = 7 - (2 + 1/2(8)) = +1$ $FC_O = 6 - (6 + 1/2(2)) = -1$ $FC_O = 6 - (4 + 1/2(4)) = 0$
Double bonded O
The FC_{Br} can be improved further by forming a second double bond to one of the other O's (structure II).
$FC_{Br} = 7 - (2 + 1/2(10)) = 0$ $FC_O = 6 - (6 + 1/2(2)) = -1$ $FC_O = 6 - (4 + 1/2(4)) = 0$
Double bonded O's

Structure II has the more reasonable distribution of formal charges.
The oxidation numbers (O.N.) are: $O.N._{Br} = +5$ and $O.N._O = -2$.



I II

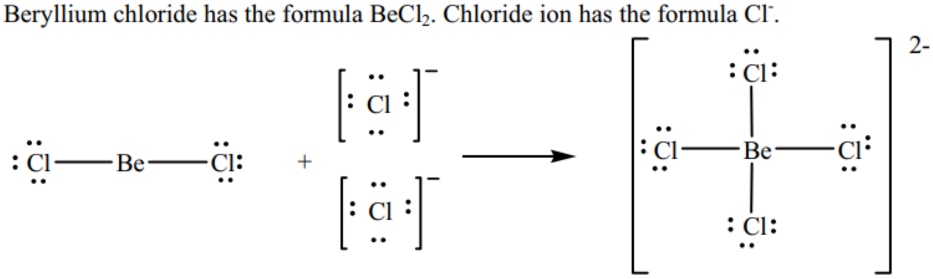
Check: The total formal charge equals the charge on the ion (-1).
b) SO_3^{2-} has 26 valence electrons.
Placing the O's around the central S and forming 3 S-O bonds uses 6 electrons and leaves 20 electrons (10 pairs). Placing 3 pairs on each O (3 x 3 = 9 total pairs) leaves 1 pair for the S and yields structure I below. In structure I all the atoms have a complete octet. Calculating formal charges:
$FC_S = 6 - (2 + 1/2(6)) = +1 FC_O = 6 - (6 + 1/2(2)) = -1$
The FC_O is acceptable, but FC_S is larger than is usually acceptable. Forming a double bond between any one of the O's gives:
$FC_S = 6 - (2 + 1/2(8)) = 0$ $FC_O = 6 - (6 + 1/2(2)) = -1$ $FC_O = 6 - (4 + 1/2(4)) = 0$
Double bonded O



I II

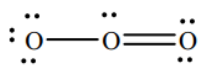
Structure II has the more reasonable distribution of formal charges.
The oxidation numbers (O.N.) are: $O.N._S = +4$ and $O.N._O = -2$.
Check: The total formal charge equals the charge on the ion (-2).

10.23 Plan: Draw Lewis structures for the reactants and products.
Solution:
Beryllium chloride has the formula $BeCl_2$. Chloride ion has the formula Cl^-.

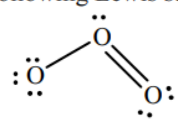


10.26 Plan: Use the structures in the text to determine the formal charges.
Solution:
Structure **A**: $FC_C = 4 - (0 - 1/2(8)) = 0$; $FC_O = 6 - (4 - 1/2(4)) = 0$; $FC_{Cl} = 7 - (6 - 1/2(2)) = 0$
Total FC = 0
Structure **B**: $FC_C = 4 - (0 - 1/2(8)) = 0$; $FC_O = 6 - (6 - 1/2(2)) = -1$;
$FC_{Cl} = 7 - (4 - 1/2(4)) = +1$ (double bonded) $FC_{Cl} = 7 - (6 - 1/2(2)) = 0$ (single bonded)
Total FC = 0
Structure **C**: $FC_C = 4 - (0 - 1/2(8)) = 0$; $FC_O = 6 - (6 - 1/2(2)) = -1$;
$FC_{Cl} = 7 - (4 - 1/2(4)) = +1$ (double bonded) $FC_{Cl} = 7 - (6 - 1/2(2)) = 0$ (single bonded)
Total FC = 0
Structure **A** has the most reasonable set of formal charges.

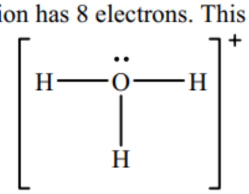
10.34 Plan: First draw a Lewis structure, and then apply VSEPR.
Solution:
a) The molecule has 18 electrons. This gives the following Lewis structure:

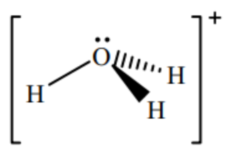


or

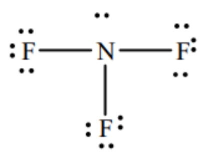


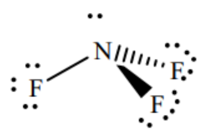
There are three groups around the central O, one of which is a lone pair. This gives a **trigonal planar** electron-group arrangement, a **bent** molecular shape, and an ideal bond angle of **120°**.
b) This ion has 8 electrons. This gives the following Lewis structure:





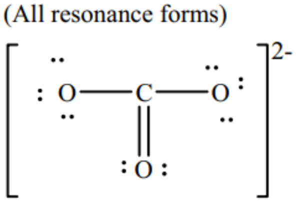
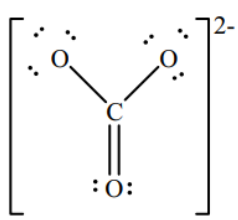
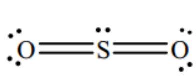
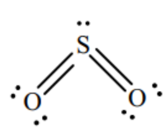
There are four groups around the O, one of which is a lone pair. This gives a **tetrahedral** electron-group arrangement, a **trigonal pyramidal** molecular shape, and an ideal bond angle of **109.5°**.
c) The molecule has 26 electrons. This gives the following Lewis structure:



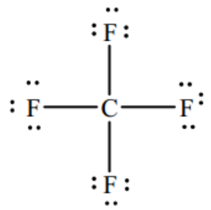
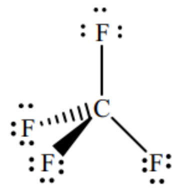


There are four groups around the N, one of which is a lone pair. This gives a **tetrahedral** electron-group arrangement, a **trigonal pyramidal** molecular shape, and an ideal bond angle of **109.5°**.

10.36 Plan: First draw a Lewis structure, and then apply VSEPR.
Solution:

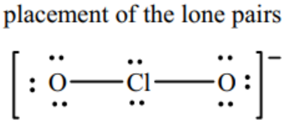
Lewis Structure	Electron-group Arrangement	Molecular Shape	Ideal Bond Angle
(a) (All resonance forms)	Trigonal Planar	Trigonal Planar	120°
 + 2 additional resonance forms			
(b)	Trigonal Planar	Bent (V-shaped)	120°
			

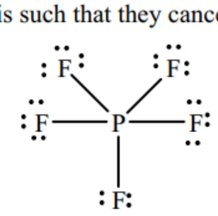
(c) Tetrahedral Tetrahedral 109.5°

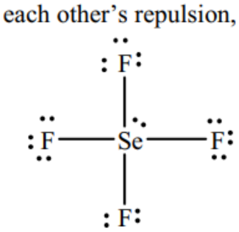
 

10.38 Plan: Examine the structure shown, and then apply VSEPR.
Solution:
a) This structure shows three electron groups with three bonds around the central atom.
There appears to be no distortion of the bond angles so the shape is **trigonal planar**, the classification is **AX_3**, with an ideal bond angle of **120°**.
b) This structure shows three electron groups with three bonds around the central atom.
The bonds are distorted down indicating the presence of a lone pair. The shape of the molecule is **trigonal pyramidal** and the classification is **AX_3E**, with an ideal bond angle of **109.5°**.
c) This structure shows five electron groups with five bonds around the central atom.
There appears to be no distortion of the bond angles so the shape is **trigonal bipyramidal** and the classification is **AX_5**, with ideal bond angles of **90°** and **120°**.

10.40 Plan: The Lewis structures must be drawn, and VSEPR applied to the structures. The structures are shown after the discussion.
Solution:
a) The ClO_2^- ion has 20 valence electrons. The Cl is the central atom. There are two bonds (to the O's) and two lone pairs on the Cl (AX_2E_2). The shape is **bent** (or V-shape). The structure is based on a tetrahedral electron-group arrangement with an ideal bond angle of **109.5°**. The presence of the lone pairs will cause the remaining angles to be **less than 109.5°**.
b) The PF_5 molecule has 40 valence electrons. The P is the central atom. There are 5 bonds to the P and no lone pairs (AX_5). The shape is **trigonal bipyramidal**. The ideal bond angles are **90° and 120°**. The absence of lone pairs means the **angles are ideal**.
c) The SeF_4 molecule has 34 valence electrons. The Se is the central atom. There are 4 bonds to the Se which also has a lone pair (AX_4E). The shape is **see-saw**. The structure is based on a trigonal bipyramidal structure with ideal angles of **90° and 120°**. The presence of the lone pairs means the angles are **less than ideal**.
d) The KrF_2 molecule has 22 valence electrons. The Kr is the central atom. There are 2 bonds to the Kr which also has 3 lone pairs (AX_2E_3). The shape is **linear**. The structure is based on a trigonal bipyramidal structure with ideal angles of 90° and 120°. The placement of the F's makes their ideal bond angle to be 2 x 90° = **180°**. The placement of the lone pairs is such that they cancel each other's repulsion, thus the actual **bond angle is ideal**.


(a)


(b)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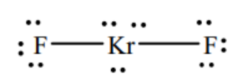
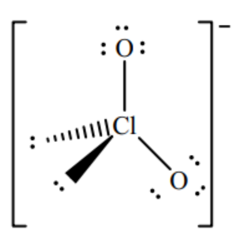
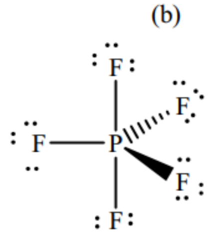
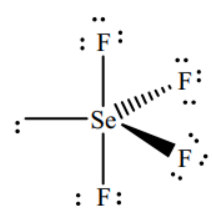
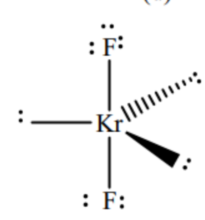

(d)


(a)


(b)


(c)


(d)

10.48 Plan: The ideal bond angles depend on the electron-group arrangement. Deviations depend on lone pairs.
Solution:
a) The C and N have 3 groups, so they are ideally 120°, and the O has 4 groups, so ideally the angle is 109.5°. The N and O have lone pairs, so the angles are less than ideal.
b) All central atoms have 4 pairs, so ideally all the angles are 109.5°. The lone pairs on the O reduce this value.
c) The B has 3 groups leading to an ideal bond angle of 120°. All the O's have 4 pairs (ideally 109.5°), 2 of which are lone, and reduce the angle.

10.55 Plan: To determine if a bond is polar, determine the electronegativity difference of the atoms participating in the bond. To determine if a molecule is polar, it must have polar bonds, and a certain shape determined by VSEPR.

Solution:

a) The greater the difference in electronegativity the more polar the bond:

Molecule	Bond	Electronegativities	Electronegativity difference
SCl_2	S-Cl	S = 2.5 Cl = 3.0	3.0 - 2.5 = 0.5
F_2	F-F	F = 4.0 F = 4.0	4.0 - 4.0 = 0.0
CS_2	C-S	C = 2.5 S = 2.5	2.5 - 2.5 = 0.0
CF_4	C-F	C = 2.5 F = 4.0	4.0 - 2.5 = 1.5
BrCl	Br-Cl	Br = 2.8 Cl = 3.0	3.0 - 2.8 = 0.2

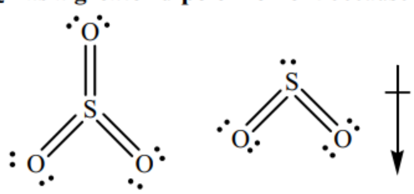
The polarities of the bonds increases in the order: F-F = C-S < Br-Cl < S-Cl < C-F. Thus, **CF_4** has the most polar bonds.

b) The F_2 and CS_2 cannot be polar since they do not have polar bonds. CF_4 is a AX_4 molecule, so it is tetrahedral with the 4 polar C-F bonds arranged to cancel each other giving an overall nonpolar molecule. **BrCl has a dipole moment** since there are no other bonds to cancel the polar Br-Cl bond. **SCl_2 has a dipole moment** (is polar) because it is a bent molecule, AX_2E_2, and the S-Cl bonds both pull to one side.

10.57 Plan: If only 2 atoms are involved, only an electronegativity difference is needed. If there are more than 2 atoms, the structure must be determined.

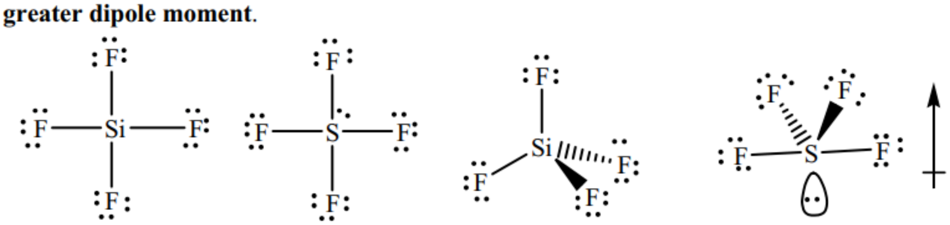
Solution:

a) All the bonds are polar covalent. The SO_3 molecule is trigonal planar, AX_3, so the bond dipoles cancel leading to a nonpolar molecule (no dipole moment). The SO_2 molecule is bent, AX_2E, so the polar bonds both pull to one side. **SO_2 has a greater dipole moment** because it is the only one of the pair that is polar.



b) ICl and IF are polar, as are all diatomic molecules composed of atoms with differing electronegativities. The electronegativity difference for ICl is less than that for IF. The greater difference means that **IF has a greater dipole moment**.

c) All the bonds are polar covalent. The SiF_4 molecule is nonpolar (has no dipole moment) because the bonds are arranged tetrahedrally, AX_4. SF_4 is AX_4E, so it has a see-saw shape, where the bond dipoles do not cancel. **SF_4 has the greater dipole moment**.



d) H_2O and H_2S have the same basic structure. They are both bent molecules, AX_2E_2, and as such, they are polar. The electronegativity difference in H_2O is greater so **H_2O has a greater dipole moment**.